

HW 12
Dr. Robertson
STAT 601

Total points 100

Packages required: Ggplot2, Likert package, and other possible: grid, and RColorBrewer

Graded on:

1. Code Quality
2. Figure Quality
3. Narrative quality
4. Conclusions on the data analyzed

Likert Scales in Survey Design

Likert scales are a method of measuring the strength of responses when deploying surveys that sometimes are categorical and can be nominal and ordinal and represents a critical examination of how survey data can be analyzed numerically. The literature review poses some ideas of whether you should use anova with numerical survey data. Read through the literature and gain an understanding of this idea.

Step 1: Review the Literature (10 points)

Read the literature Review and create a $\frac{1}{2}$ to $\frac{3}{4}$ page analysis of the nature of likert scales and comment on the following with your data:

- Explain the nature of likert scales, its origins, methods and possible parametric tests
- Is the data in this exercise ordinal or nominal
- Could you use anova on these questions from the survey, why or why not
- Finally, comment on if parametric test could be part of the exploratory data analysis

The Brief

Suppose a survey was deployed to study the sentiment of students' view of generative AI. The study was interested looking at different responses from students about a variety of feelings about AI.

The survey was coded by number but are actually responses from the students that answered:

"Strongly Disagree", "Disagree", "Undecided", "Agree", "Strongly Agree"

1	2	3	4	5
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The questions in the survey are as follows:

- Q1 I usually use AI in my studies
- Q2 I think AI is unethical

- Q3 I think AI should be used in my Homework unconditionally
- Q4 I choose not to disclose my AI use
- Q5 I feel AI is going in the right direction

Step 2: Perform an exploratory data analysis and draft a preliminary report about the data structure: (10 points)

Report on:

- The distribution of the gender and class of samples taken
- Identify any outliers or missing data and explain how you would handle the missing data
- Examine the nature of each response from each question and comment preliminarily what you can see about the class and gender as it relates to the nature of the responses
- To create the next steps, you may need to omit data you have reported on, so make sure to decide and report

Step 3: Use GGplot2 to create a set of diagrams that show the distribution of: (10 points)

1. Gender
2. Class
3. Q1-Q5

Make sure these plots are professional and report information as if you were drafting a report for a consulting client. For full credit these figures need to have a title, legend, axes labels, and use colors.

Step 4: Report on your GGplot2 results and what strategies that you might employ further study the structure of the data and how this could assist in Step 5's reporting. (10 points)

Step 5: Clean and parse the data as follows: (5 points)

1. Make sure you create a set of factors that replaces the numbers with the responses in the dataset

"Strongly Disagree",	"Disagree",	"Undecided",	"Agree",	"Strongly Agree"
1	2	3	4	5
2. Change and reorder the class factors if necessary so freshman is first and senior is last
3. Check and make sure all levels are represented
4. Look at the order of the levels such that class and order of the factors above are in the right order
5. Check the levels of each likert they must be identical, as you will see Q4 has missing factor so we will need to code that in to make sure everything works (use the levels() to verify if a factor is missing, and if so you need to add that factor to the question. ***This will be the most challenging part is to add an additional factor correctly so it is a placeholder so Q4 will have all of the correct factors.***

The way to create the final tables for analysis is:

1. Every questions must have the same number of levels this is important
2. You have to decide if the missing values also create an error when trying render the figures using the likert package
3. Do some tests in the package to play around with how certain conditions should be met before making the tables

Step 6: Create the 3 types of plots the likert package allows for: (30 points)

1. Bar type
2. Heat type
3. Density type

There should be 3 plots for each A, B, and C. They will be the same plots but with different metrics.

- A. Create a Bar, heat and density plot that has all of the questions and correct “strongly disagree” “strongly agree” in each of the plots and make sure you use labs to create a title and use plot.title, theme, and other code options to create a well designed visual of the questions so we can study the results. Comment on what you see in each plot and analyze the results. All questions need to be included in all plots for comparison.
- B. Split the data by gender and produce separate plots of the survey results, there will 3 for male and 3 for female. Conduct an analysis and report on what you see from the results of the 6 tables
- C. Split the data by class and produce the 3 plots that show the differences among classes and their responses to the question, here we have 4 classes (fresh,soph,jun,sen) so you will have 12 plots to report on and conduct an analysis of how each class answered the different questions of AI survey

Step 6: After you have produced all of the tables conduct a full analysis of all of these results. This step is expected to be a professional write up with using the plots produced above you will graded one the follow: (25 Points)

1. The length and depth of analysis should not exceed 4 additional pages above and beyond what you have already done.
2. Pick some highlights about the result and show your strength of insights as to the results of the survey
3. You must cover the overall results from 5A, 5B, and 5C
4. Find some way to show these comparisons that gives the reader a good look of what you have found
5. Finally develop a solid conclusion about what you have found and whether we have gained insight into students use of AI

The quality of this report is what is most important, this is to be a professional write up, not homework type where you type number give the answer to hand in. This is one the final topics we will cover and so you will be heavily docked if you do not produce a quality analysis.